ZA COLLECTIBLES

Website Design & Architecture Specification

Behavioural-value design, uncertainty-first assessment and modular collectible intelligence

> **Core design law: the site must feel expensive where status and acquisition value are created, and feel effortless where evidence capture and uncertainty reduction create value. Do not optimise the entire platform for one psychological regime.**

Project status: Initial live build

Initial live lanes: Raw loose coins and certificate-backed gemstones

Source basis: ZAC Collectible Intelligence Platform - Project Blueprint v2, plus the behavioural design direction supplied for this website specification.

Version: Website Design & Architecture v1

Document Map

1. Executive Design Thesis

The ZA Collectibles website must not be designed as a generic online shop with an appraisal form bolted onto it. It is a two-sided value system. On one side, it converts imperfect evidence into structured provisional intelligence. On the other, it presents selected collectibles as assets with history, scarcity, status and provenance. The same brand must support both functions, but the user psychology is different.

The project blueprint already establishes the operating law: incomplete evidence is a normal input state, better evidence should tighten conclusions rather than determine whether the system works at all, and every output should guide the user towards the next highest-value action. The website must make that law visible in its interface rather than burying it in backend logic.

> **The mistake to avoid is applying 'luxury friction' everywhere. Prestige friction belongs at the acquisition boundary. Assessment intake should remove anxiety and effort until the system has created enough value to justify asking for more.**

The design objective is therefore not minimum clicks. It is maximum perceived and practical value per interaction. Every screen should answer one of four questions: What is this? Why should I trust it? What is it worth in this context? What should I do next?

Design outcomes

- Create immediate category authority without imitating a jewellery retailer, auction house or mass-market coin dealer.
- Make uncertainty legible: observed facts, inferences, missing evidence, confidence and next actions must be visually distinct.
- Use visual atmosphere and semantic framing to increase perceived seriousness, provenance and asset character.
- Preserve low-friction pathways for casual owners who begin with weak evidence.
- Create deliberate, status-compatible friction for high-value acquisition, specialist review and exceptional items.
- Keep the frontend category-specific while maintaining one reusable backend workflow, data model and registry.
- Ensure every interaction generates reusable structured data rather than disposable correspondence.

2. The Central Synthesis: Two Psychological Regimes

A Sutherland-inspired design approach should not be interpreted as 'make everything slower, darker and harder'. The more useful interpretation is to recognise that efficiency is not synonymous with value. Different moments in the journey require different signals.

Regime	Primary user state	Desired psychological signal	UX stance
Assessment / intelligence	Curiosity, uncertainty, anxiety, incomplete evidence	Competence, helpfulness, progress, intellectual honesty	Low-friction entry; progressive disclosure; clear next action
Acquisition / vault	Desire, comparison, status evaluation, willingness to spend	Scarcity, seriousness, provenance, selectivity, institutional confidence	Deliberate pacing; abundant space; selective information; controlled transaction
Escalation / specialist	Risk, conflicting evidence, high value or suspicion	Judgement, discipline, procedural seriousness	More evidence required; explicit hold states; human review
Registry / returning owner	Accumulated ownership, portfolio management	Continuity, stewardship, growing intelligence	Persistent records; history; re-assessment; evidence upgrades

Behavioural rule for friction

Friction is justified only when it changes meaning, improves evidence quality, protects trust or increases perceived selectivity. Friction that merely wastes time is not prestigious; it is incompetent.

- Reduce friction before the user has received value.
- Introduce effort when better evidence materially narrows identification, authenticity, condition or value uncertainty.
- Use deliberate pacing around high-ticket objects to prevent the interface from feeling transactional and disposable.
- Reserve application-style interaction for specialist appraisal, consignments and exceptional acquisitions, not for ordinary enquiries.
- Never hide basic navigation, accessibility or security behind theatrical design.

3. Experience Architecture and Site Map

The information architecture should separate the two major intents from the first screen: 'Assess what I have' and 'Explore what is available'. Do not force assessment users through retail inventory and do not force collectors through an appraisal funnel.

Top-level area	Purpose	Primary actions
Home	Establish authority, explain the two propositions and set a high value anchor	Assess an item; enter The Vault; view a featured object

Top-level area	Purpose	Primary actions
Assess	Uncertainty-first intake and provisional assessment	Snapshot; Quick Capture; Structured Capture
The Vault	Curated acquisition environment	Browse curation; inspect provenance; request acquisition
Disciplines	Category knowledge architecture	Numismatics; Gemmology; later Militaria, Horology and other modules
Reports / Registry	Return-user intelligence workspace	View reports; add evidence; request update; compare revisions
About / Method	Explain how the platform reasons and where it stops	Read methodology; confidence model; limitations
Specialist Review	Escalated or high-value path	Submit deeper evidence; request human review; initiate provenance work

Recommended primary navigation

ASSESS | THE VAULT | DISCIPLINES | METHOD | REGISTRY

Use 'The Vault' as a controlled curation area rather than as a synonym for every object in the database. The Registry is the intelligence layer. The Vault is the acquisition layer. Conflating them destroys both meanings.

4. Behavioural-Value Design Doctrine

The behavioural layer should be treated as a design system, not as decorative copy. The following principles translate the project's Sutherland-oriented direction into repeatable interface rules.

Principle	UI/UX directive
Context creates value	Objects must appear in a setting that signals curation and seriousness. Avoid commodity-grid treatment for high-value material.
Costly signalling	Use restraint, large visual fields, strong photography and editorial pacing. Do not simulate cost with gratuitous animation or visual clutter.
Anchoring	Lead with one genuinely exceptional object or assessment case. The anchor must be credible, not fabricated.
Choice architecture	Reduce undifferentiated options. Route users by intent, evidence state and value context rather than by enormous category menus.
Effort justification	Where deeper capture is required, explain what additional certainty the effort buys. The effort becomes meaningful rather than bureaucratic.
Scarcity and selectivity	Use factual availability states such as 'currently under review', 'private acquisition discussion' or 'one known example in current curation' only when true.
Semantic reframing	Use asset, curation, provenance and assessment language where accurate. Avoid retail language that collapses differentiated objects into SKUs.
Asymmetric value	Prioritise small changes that alter perception strongly: photography, naming, sequencing, confidence visualisation and report ritual can add more value than feature volume.
Ritual	The report reveal, acquisition request and registry update should feel like consequential events, not ordinary form submissions.
Honest ambiguity	Do not manufacture precision. A well-explained range with visible confidence can feel more expert than a false single number.

5. Visual Identity and Atmosphere

The visual system should communicate quiet institutional confidence. It should borrow from museum exhibition, archival documentation and private banking rather than from conventional e-commerce.

5.1 Colour

- Default public presentation: deep charcoal or slate surfaces for acquisition, editorial and hero contexts.
- Assessment forms may use a lighter neutral work surface for readability, document comparison and image inspection.
- Use a restrained metallic or antique-gold accent only for hierarchy, verified states, provenance markers and selected actions.
- Avoid bright retail colours, promotional red, neon status badges and coupon-style contrast.
- Reserve red for genuine blockers, mismatch risk or critical escalation.

5.2 Typography

Use an authoritative serif for display headings and object titles, paired with a highly legible sans-serif for controls, data and body text. Typography should signal heritage without becoming faux-antique.

Role	Recommendation	Rule
Display / editorial	Garamond-family, Baskerville-family or similarly restrained serif	Large, sparse, short lines
UI / data	Inter, Source Sans, Helvetica/Arial-family	High legibility; tabular numerals for measurements and ranges
Certificates / evidence	Monospace only where machine-like data comparison helps	Never use monospace as a decorative 'forensic' effect

5.3 Photography and media

- Hero objects: cinematic macro stills or slow video pans with controlled highlights and realistic shadows.
- Coins: obverse, reverse and edge views should become navigable evidence, not just a carousel.
- Gemstones: front, side, inscription and certificate views should support side-by-side comparison.
- Use texture: strike detail, inclusions, patina, edge wear, certificate embossing and surface marks should remain visible.
- Never over-retouch defects that influence identification, authenticity, condition or value.
- Use neutral evidence photography inside reports even if the public editorial layer uses dark atmospheric imagery.

5.4 Space, density and motion

High-value public pages should use abundant negative space and slow visual cadence. Data-rich assessment pages should be denser. This is an intentional contrast: museum outside, analyst workstation inside.

- No autoplay sound.
- Video motion should be slow, optional and immediately pausable.
- Do not use parallax, scroll-jacking or elaborate transitions in evidence-capture flows.
- Use motion to reveal hierarchy or state changes, not to manufacture luxury.

6. Global UI System and Interaction Grammar

The interface should use a small, disciplined component vocabulary.

Component	Use	Behavioural purpose
Primary action	Assess, Continue, Request Acquisition	One dominant next move
Secondary action	Add evidence, Save for later, View method	Preserve optionality without visual competition
Confidence band	Identification, authenticity, condition, value	Make uncertainty concrete
Evidence chip	Photo present, certificate verified, measurement missing	Expose what conclusion rests on
Blocker card	Mismatch, suspected alteration, missing value-sensitive detail	Turn caution into visible expertise
Next-action card	Take edge photo, verify report number, request specialist review	Direct effort to highest-value information
Provenance timeline	Ownership, source, certificate, prior valuation, revisions	Convert history into structured value
Valuation context selector	Collector, dealer, liquidation, insurance, auction, estate	Prevent generic pseudo-pricing
Revision marker	Assessment v1, v2 etc.	Show that intelligence improves with evidence

Status language

State	Preferred wording	Avoid
Thin evidence	Early-stage assessment	Insufficient data
Wide range	Low-confidence provisional range	Unknown value
Missing detail	Value-sensitive detail not yet visible	Incomplete submission
Need more media	Further capture recommended	Upload required
Escalation	Specialist review recommended / assessment held for review	System failure
Verified document	Certificate data verified against issuing source	Authentic stone
Mismatch	Certificate-to-item consistency concern	Fake

7. Homepage and First-Impression Architecture

The homepage should establish value before explaining features. It should not open with a product grid, discount language or a generic 'buy and sell collectibles' proposition.

Above the fold

1. Featured anchor object: one exceptional coin, gemstone or historically significant collectible presented full-bleed or in a large editorial frame.
2. One-line positioning statement: structured collectible intelligence and selective acquisition, not mass-market retail.
3. Two dominant routes: 'Assess what I have' and 'Enter The Vault'.
4. One quiet trust cue: methodology, verified provenance practice, specialist escalation or evidence-aware reporting.

The featured anchor should create a high reference point for the entire site. It must be real and defensible. Do not use arbitrary strikethrough pricing or fabricated 'worth' comparisons.

Homepage sequence

- Hero anchor object
- Two-route intent choice
- How assessment works: submit -> provisional intelligence -> improve certainty
- Selected current curation: maximum three to six objects, not an endless grid
- Example report fragment showing range, confidence and next action
- Disciplines / areas of expertise
- Method and limitations
- Invitation to specialist review or acquisition discussion

8. Assessment Intake: Snapshot, Quick and Structured Capture

The assessment path implements the uncertainty-first operating model. The user must be allowed to begin with whatever they have and progressively improve the evidence packet.

8.1 Capture mode selector

Mode	What the UI asks for	What the user gets	Commercial role
Snapshot	One or a few broad photos	Category guess, visible highlights, obvious risk areas, rough band and next best photos	Lead generation and curiosity conversion
Quick Capture	Basic item photos, simple notes or rough counts	Probable identification, rough value band, visible condition notes and capture recommendations	Fast triage / semi-automated tier
Structured Capture	Photos, measurements, certificate data, provenance and valuation purpose	Tighter identification, stronger range, confidence breakdown and blocker logic	Core paid assessment
Escalation	Specialist evidence, conflicting data or high-value case	Escalation rationale, evidence gaps, specialist route and hold decision	Trust protection and premium service

8.2 Progressive capture mechanics

- Show required fields only for the current mode. Do not expose the full schema at entry.
- After each upload, run lightweight completeness/blocker logic and show what new evidence would change the conclusion most.
- Use visual capture guides over text-heavy instructions where possible.
- Allow 'I don't know' for measurements, metal, origin and prior value. Unknown is a valid state.
- Save partial submissions automatically.
- Let users upgrade a Snapshot into Quick or Structured Capture without re-entering existing data.
- On mobile, prioritise camera capture, crop guidance and sequence control.

8.3 Evidence-value explanation

Every request for more effort should state the expected payoff. Example: 'An edge photo can materially change identification confidence for this coin.' That language converts a form requirement into an intelligence transaction.

9. Category-Specific Capture Modules

9.1 Raw loose coins

Minimum viable intake:

- One or more clear photographs, ideally front and back
- Rough owner note
- Optional country, date or denomination if known

Full intake:

- Obverse, reverse and edge photographs
- Weight, diameter and thickness
- Scale or ruler photograph
- Metal if known
- Provenance note
- Close-ups of unusual, damaged or cleaned areas

Coin-specific UI states:

- Cleaning suspected
- Damage materially affects value
- Counterfeit / alteration concern
- Measurement conflict
- Likely grade band only
- Variety attribution not supportable from current evidence

9.2 Certificate-backed gemstones

Minimum viable intake:

- Certificate image or PDF
- Visible report number where possible
- One clear stone photograph
- Stated use case if known

Full intake:

- Certificate file and report number
- Front and side stone photographs
- Inscription photograph where available
- Dimensions and carat weight
- Invoice or prior valuation
- Origin note
- Intended valuation purpose

Gemstone-specific UI states:

- Certificate extracted
- Certificate source verification pending / verified / failed
- Photo-to-certificate consistency: strong / partial / conflict
- Inscription observed / not visible / inconsistent
- Natural, lab-grown or simulant status from certificate data
- Treatment disclosure present / absent / unclear

10. Assessment Report and Guidance Interface

The digital report is the principal commercial product. It should feel like a structured intelligence brief rather than a generated answer. The user must be able to see the reasoning boundary, confidence and next action at a glance.

10.1 Report header

- Item working title and item ID
- Assessment date and revision
- Selected valuation context
- Current value range or band
- Overall state: provisional / review recommended / held for escalation
- One-sentence conclusion

10.2 Confidence dashboard

Display four separate confidence measures: identification, authenticity, condition and value. Do not compress them into one score because the failure modes differ.

Confidence dimension	Question answered	Example evidence drivers
Identification	How likely is the item correctly identified?	Visible legends, measurements, certificate fields, date, mint, shape
Authenticity	How comfortable are we that it appears genuine or matched?	Surface evidence, certificate verification, inscription, contradictions
Condition	How reliable is the condition judgement?	Image quality, angles, visible damage, cleaning, wear
Value	How dependable is the range for the selected purpose?	Identification, authenticity, condition, comparable evidence freshness

10.3 Observed / inferred / missing

Every report should separate three epistemic states visually. This is critical to the platform's credibility.

State	Meaning	UI treatment
Observed	Directly visible or extracted from submitted evidence	Neutral factual rows
Inferred	Reasonable interpretation from observed evidence	Labelled hypothesis / probable identification
Missing	Value-sensitive evidence not yet available	Actionable gap with next capture step

10.4 Valuation context

Do not show a value band until a valuation purpose is selected or explicitly defaulted. The selector should explain the meaning of each range in plain language.

- Collector market range
- Dealer purchase range
- Liquidation range
- Insurance replacement support
- Auction estimate
- Estate or probate reference

10.5 Guidance engine

Every report ends with one primary next action and up to two secondary actions. The primary action should be the step with the highest expected reduction in uncertainty or highest commercial payoff.

> **Example: 'Next best action: photograph the coin edge in diffuse light. This can materially distinguish two candidate varieties and may narrow the value range.'**

11. The Vault: Acquisition and High-Value Inventory Experience

The Vault is not a standard catalogue. It is a curated acquisition environment. Browse behaviour may remain efficient, but the presentation should resist the visual language of commodity e-commerce.

11.1 Catalogue architecture

- Default view: editorial cards with large imagery and sparse metadata.
- Do not show endless 4- or 5-column grids on desktop.
- Use 1-2 columns for exceptional items, 2-3 for ordinary curated stock.
- Filters remain available but visually subordinate: discipline, period, origin, material, availability, acquisition range.
- Sort language: Recently Curated, Historical Significance, Rarity / Scarcity, Acquisition Range. Avoid 'Best Sellers'.
- Where pricing is public, display it quietly. Do not use discount badges or urgency timers.

11.2 Transaction controls

Use transaction language that reflects the type and value of the object. Not every item needs identical friction.

Item tier	Primary CTA	Price treatment	Friction
Accessible / lower-value	Acquire / Reserve	Public price	Conventional checkout permitted
Curated mid-tier	Request Acquisition	Public or guide price	Identity/contact + acquisition discussion
High-value / exceptional	Request Private Acquisition	POA or qualified range	Application-style contact; provenance acknowledgement; specialist handling
Not for immediate sale	Register Interest	No price	Waitlist / future disposition only

Price on application should be selective, not universal. Hiding every price creates suspicion and suppresses qualified demand. Use POA where the object, transaction complexity or negotiation context genuinely benefits from it.

12. Item Detail and Provenance Architecture

An object page must answer more than 'what is it and how much does it cost?'. It should convert physical and documentary evidence into a coherent value narrative.

Recommended item page sequence

1. Hero macro image / controlled media
2. Object title, period, discipline and availability state
3. Acquisition action
4. Why this object matters: concise historical or collector significance
5. Evidence gallery: front/back/edge, inscriptions, marks, certificate
6. Provenance timeline
7. Condition and known interventions
8. Authentication / certificate state
9. Dimensions and technical data
10. Comparable context or valuation basis where appropriate
11. Acquisition process and logistics
12. Related objects by discipline or provenance, not 'customers also bought'

Provenance as a first-class object

Provenance should not be a paragraph buried below specifications. It should be a structured timeline capable of storing source, date, owner, document, confidence and supporting evidence. This same model should feed public display, internal registry and future due diligence.

13. Language, Semantic Framing and Microcopy

Interface language should change perceived category without becoming pretentious or misleading. Use precise specialist vocabulary when it communicates real distinctions.

Commodity wording	Preferred wording	Use condition
Shop	The Vault / Current Curation	Acquisition area
Products	Objects / Pieces / Items	General collectible inventory
Categories	Disciplines	Top-level scholarly or collecting domains
Add to cart	Acquire / Request Acquisition	Based on transaction tier

Commodity wording	Preferred wording	Use condition
Contact us	Request Review / Acquisition Enquiry / Specialist Review	Intent-specific
Upload more	Add evidence	Assessment flows
Result	Assessment	Provisional intelligence output
Price	Acquisition price / valuation range	Match actual commercial meaning
Out of stock	Acquired / No longer available	Unique or non-repeatable objects
Account	Registry	Returning owner / assessment workspace

Do not misuse elevated terminology. 'Numismatics' is appropriate as a discipline. It does not replace 'coin' everywhere. Good semantic framing raises perceived competence because it is accurate, not because it is obscure.

14. System Architecture

The frontend should remain a thin orchestration and presentation layer over a modular intelligence backend. The technical architecture must support incomplete evidence, asynchronous processing, revision history and category plug-ins.

Layer	Responsibilities	Implementation directive
Client UI	Capture, browsing, report display, registry, acquisition interactions	Responsive web app; componentised category modules
API / application layer	Auth, sessions, case state, permissions, workflow orchestration	Versioned API; idempotent submission and retry logic
Media service	Image/PDF upload, transformation, thumbnails, metadata	Object storage; signed URLs; immutable originals
Intake normaliser	Convert uploads and notes into a clean packet	Schema-constrained output; preserve raw input
Assessment services	Coin ID/risk, gem extraction/consistency, ranging	Modular services; no single monolithic prompt
Confidence engine	Four confidence dimensions	Store score, rationale and evidence dependencies
Guidance engine	Rank next best actions	Action value based on expected uncertainty reduction / commercial relevance
Registry	Persistent item, collection, evidence, report and revision records	Relational core + object storage references
QA / review	Adversarial review, blocker enforcement, release decisions	Human-review queue with explicit release rules
Market evidence service	Comparable and pricing evidence	Source-date logging, provenance and revalidation
Acquisition service	Interest, reservation, offer/request states	Separate from assessment valuation records

14.1 Recommended event flow

UPLOAD -> NORMALISE -> COMPLETENESS/BLOCKER CHECK -> CATEGORY MODULE -> CONFIDENCE -> VALUATION MODE -> RANGE -> QA -> REPORT -> REGISTRY -> GUIDANCE / ESCALATION

Expensive or deep model calls should not run before basic completeness and contradiction checks. The workflow should terminate early into a useful output when evidence is weak, rather than failing.

14.2 Frontend state model

- Draft intake
- Submitted / normalising
- Needs additional evidence
- Assessment in progress
- Provisional assessment ready
- QA review
- Released
- Escalated / on hold

- Updated evidence received
- Re-assessment in progress
- Superseded by later revision

15. Shared Data Model, State Model and Registry

The top-level data model remains category-agnostic. Every object should pass through intake, interpretation, assessment, valuation, QA, report and registry states. Only category-specific fields vary.

Field group	Required shared fields
Identity	Item ID, Collection ID, Client/Owner, Category, Subcategory, Working title
Valuation context	Purpose, requested output, urgency, intended use, commercial stage
Source material	Photo set, file set, certificate/document, notes, measurement status
Assessment control	Assessor, AI run version, review status, assessed date, revision
Confidence	Identification, authenticity, condition, value
Risk and blockers	Blocker flags, escalation flag, warning notes, required follow-up, unresolved contradictions
Output state	Provisional description, findings, value range, next action, report link, registry status

Registry requirements

- Immutable original uploads
- Derived/cropped images stored separately
- Every report linked to the exact evidence set used
- Every AI run version and prompt/module version logged
- Confidence rationale stored, not only displayed score
- Market evidence source and date logged
- Revision history visible to internal users and selectively visible to clients
- Ability to merge duplicate item records without losing history
- Collection-level grouping and portfolio view

16. Prompt Orchestration and AI Service Hooks

The blueprint's modular prompt stack should be implemented as explicit services with typed inputs and outputs. Do not allow a single free-form prompt to own the entire workflow.

Module	Job	Required UI/system hook
P1 Intake normaliser	Raw uploads -> clean intake packet	Runs immediately after submission
P2 Completeness/blocker checker	Missing media/fields and obvious contradictions	Feeds evidence-gap UI before deeper work
P3 Coin identifier	Candidate coin identity from visuals/measurements	Feeds probable identity, alternatives and confidence
P4 Coin condition/risk	Wear, cleaning, damage, authenticity concerns	Feeds condition and blocker cards
P5 Gem certificate extractor	Document -> structured fields	Auto-populates certificate pane
P6 Gem consistency checker	Compare certificate, photos and inscription	Feeds verification/mismatch state
P7 Valuation-mode selector	Apply purpose logic	Bound to user-selected valuation context
P8 Range builder	Provisional range/band	Must cite current evidence state and source-date context internally
P9 Report generator	Client report + internal note	Renders from structured record, not raw model prose
P10 Adversarial QA	Challenge assumptions and blocker handling	Required before release for configured tiers

Prompt-output discipline

- Use JSON/schema-constrained outputs for machine-critical fields.
- Store narrative explanation separately from canonical structured values.
- Require explicit 'observed', 'inferred', 'missing' classification where applicable.
- Require contradiction arrays rather than hiding conflicts inside prose.
- Never let the range builder silently overwrite category or certificate facts.
- All confidence scores need rationale and evidence dependencies.
- Failed parsing should degrade gracefully to manual fields and an early-stage assessment state.

17. Admin, QA, Escalation and Release Controls

The admin interface should function as an intelligence review console. Its purpose is not to mirror the client UI. It must expose assumptions, conflicts and release risk.

Admin case view

- Raw evidence viewer and structured record side-by-side
- Observed / inferred / missing breakdown
- All four confidence scores with rationale
- Blocker taxonomy and escalation reason
- Certificate or document extraction with source image coordinates where possible
- Comparable / market evidence with dates
- AI run and prompt-module versions
- Revision diff
- Release / return-for-evidence / escalate controls
- Internal QA note: weak assumptions, contradictions and release rationale

Mandatory release rules

- No report leaves the system without an explicit valuation purpose or an explicit default.
- No report leaves the system without all four confidence fields.
- If a blocker materially affects value, the report must state it and value confidence must fall.
- Suspected counterfeit, alteration or certificate inconsistency forces escalation.
- Every output must include the next highest-value action.
- Raw intake and structured record must remain preserved with each report version.

18. Trust, Risk, Legal and Disclosure Surfaces

Limitations should be visible at the decision point where they matter, not hidden in a footer. Trust rises when the system states exactly what it can and cannot support.

- Loose coin authenticity cannot be guaranteed from ordinary consumer photographs alone.
- Coin grading precision and rare variety attribution may require specialist review and better imaging.
- A gemstone grading certificate is not a valuation.
- Provisional ranges are purpose-specific and not interchangeable across collector, dealer, liquidation, insurance, auction and estate contexts.
- Pricing evidence ages; source dates must be logged and revalidated.
- Certificate verification confirms document/report status where available; it does not independently prove that the physical stone matches the report unless consistency evidence supports that conclusion.

Disclosure placement

Context	Disclosure treatment
Snapshot result	Short inline caveat + next capture action
Structured report	Specific limitation adjacent to affected confidence field or value range
High-value acquisition	Provenance/authentication basis clearly separated from sales narrative
Certificate-backed gemstone	Certificate status and photo-match status shown as separate states
Legal footer	Terms, privacy, data use and non-reliance language; not a substitute for contextual warnings

19. Analytics, Behavioural Experimentation and Calibration

Optimisation should measure value creation, not merely conversion rate. Some high-status design choices may reduce raw clicks while improving qualified demand, average case quality or acquisition value.

Core metrics

- Snapshot start -> completion rate
- Snapshot -> Quick / Structured upgrade rate
- Evidence request completion rate
- Time to first useful output
- Confidence improvement per additional evidence action
- Escalation rate and escalation correctness
- Report release rate after QA
- Acquisition enquiry quality and close rate
- Average acquisition value by presentation treatment
- Return rate to Registry and evidence-upgrade rate
- Percentage of valuation ranges later recalibrated materially

Behavioural experiments

Run controlled tests on framing, sequence and presentation without compromising factual truth.

Experiment	Variants	Measure
Hero anchor	Exceptional object first vs generic brand proposition	Qualified Vault exploration, enquiry quality
CTA language	Buy / Acquire / Request Acquisition	Click quality, downstream conversion, average value
Price visibility	Public price vs guide range vs POA on appropriate tiers	Qualified enquiry rate, abandonment, negotiation efficiency
Capture effort framing	Plain requirement vs 'what this evidence buys you'	Evidence completion rate
Confidence display	Numeric percentage vs band + explanation	Comprehension, trust, follow-up behaviour
Report reveal	Immediate value band vs conclusion after evidence summary	Perceived credibility, retention, next-action completion

20. Responsive Design, Performance and Accessibility

Prestige does not justify poor performance or inaccessible interaction. The site must feel deliberate, not slow.

- Optimise hero media aggressively; use poster images and adaptive video delivery.
- Lazy-load deep image galleries but preload the next likely evidence image.
- Mobile assessment capture is a primary workflow, not a reduced desktop version.
- All dark-mode text must meet contrast requirements.
- Do not encode confidence or blocker status by colour alone.
- Keyboard navigation and visible focus states remain mandatory.
- Provide alt text for public editorial imagery and functional descriptions for evidence images where appropriate.
- Allow users to pause video and disable motion.
- Use semantic HTML for report headings, tables and controls.
- Preserve readable type sizes; luxury aesthetics must not depend on tiny text.

21. Build Priorities and Implementation Sequence

The website build should track the blueprint's functional-first sequencing while adding the behavioural presentation layer in the correct places.

Phase	Build	Definition of done
1. Backbone	Shared schema, IDs, evidence storage, valuation modes, confidence model, blocker taxonomy	A blank item can enter and move through the workflow consistently
2. Intake	Snapshot, Quick and Structured capture for both live lanes	Any user can submit imperfect evidence and receive a valid state
3. Report shell	Confidence, observed/inferred/missing, range, blockers, next action	A structured report can render from canonical data
4. Coin lane	Coin prompts, capture guidance, risk states	Loose coin processes end-to-end
5. Gem lane	Certificate extraction, verification state, photo-match logic	Certificate-backed gemstone processes end-to-end

Phase	Build	Definition of done
6. Registry / QA	Persistent records, revisioning, reviewer console, release controls	Every case is reviewable and improvable
7. Vault	Curated browse, object detail, provenance and acquisition workflow	High-value inventory has a non-commodity presentation and controlled transaction path
8. Hardening	Pilot calibration, analytics, performance, accessibility, copy refinement	Public launch criteria met

Immediate build order

1. Master schema pack
2. Capture-mode framework
3. Report component system
4. Coin intake + coin prompt stack
5. Gemstone intake + certificate prompt stack
6. Registry and QA
7. Vault object-page template
8. Pilot dataset and calibration
9. Behavioural experiments

22. Acceptance Criteria

The first public deployment is acceptable only when the following are true:

- A user can submit at least one loose coin or certificate-backed gemstone and receive a structured provisional assessment.
- The system accepts thin evidence without dead-ending the user.
- The report states valuation purpose, range or band, all four confidence fields, blockers and next best action.
- Observed facts, inference and missing evidence are visually separable.
- Suspicious or conflicting cases escalate instead of producing overconfident output.
- Every case creates or updates a persistent registry record.
- Certificate-backed gemstones display certificate verification and item-to-certificate consistency as separate concepts.
- The Vault does not present high-value objects as a commodity grid or use discount-led retail conventions.
- High-value acquisition uses deliberate but justified friction.
- The site performs acceptably on mobile and low-bandwidth conditions despite rich imagery.
- The admin interface can inspect raw evidence, structured fields, QA rationale and revision history.
- All core limitations appear contextually where they affect the conclusion.

Appendix A. Canonical Interface Language

Function	Preferred label
Assessment entry	Assess What I Have
Low-friction entry	Snapshot Assessment
Intermediate capture	Quick Capture
Full capture	Structured Capture
Acquisition area	The Vault
Top-level categories	Disciplines
User-owned record area	Registry
High-value transaction	Request Acquisition
Exceptional / private transaction	Request Private Acquisition
Specialist path	Request Specialist Review
More evidence	Add Evidence
Report update	Reassess with New Evidence

Function	Preferred label
Mismatch	Consistency Concern
Wide range	Low-Confidence Provisional Range
Thin evidence	Early-Stage Assessment
Unavailable detail	Value-Sensitive Detail Not Yet Visible
Action prompt	Next Best Action

Appendix B. Page-by-Page Component Checklist

Home

Anchor object; Two intent routes; Assessment method; Selected curation; Report fragment; Disciplines; Trust / limits.

Assess landing

Mode selector; Example outputs; Evidence effort explanation; Privacy / upload note.

Snapshot capture

Camera/upload; Optional notes; Auto-save; Immediate next evidence prompt.

Quick capture

Item gallery; Basic fields; Counts/notes; Progressive evidence requests.

Structured coin

Obverse/reverse/edge; Measurements; damage/cleaning close-ups; provenance; valuation purpose.

Structured gem

Certificate; report number; stone images; inscription; measurements; prior valuation; purpose.

Report

Conclusion; value range; four confidence fields; observed/inferred/missing; blockers; next action; revision.

Registry

Collections; items; report history; evidence gaps; reassessment action.

Vault

Curated cards; filters; availability; discipline navigation.

Item detail

Macro media; significance; provenance; evidence; condition; technical data; acquisition CTA.

Specialist review

Case summary; why escalated; evidence requested; review status.

Admin case

Raw evidence; structured data; confidence rationale; blockers; comparables; AI versions; release controls.